

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Feature Type	Method Applied	Features Processed
Categorical Variables	Label Encoding (LabelEncoder)	Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, Over18, OverTime

- **Implementation Detail:** All 9 categorical columns identified as object data types were transformed into numeric representations using LabelEncoder() within a iterative loop.

2.2 Feature Transformation & Treatment

Outlier Handling (Capping)

- **Technique Applied: IQR-Based Outlier Capping** (cap_outlier function).
- **Method:** Calculated Interquartile Range Limits:
 - Lower Bound (LB)} = $Q1 - 1.5 * IQR$
 - Upper Bound (UB)} = $Q3 + 1.5 * IQR$
- **Action:** Applied series.clip(LB, UB) across all numerical features (num) to cap extreme values at upper and lower boundaries without dropping rows.

Feature Scaling

- **Technique Applied: Standardization** (StandardScaler).
- **Scope:** Applied to all input predictor variables in x_train and x_test.
- **Implementation Detail:** Standardized feature distributions to zero mean and unit variance using scaler.fit_transform(x_train) and scaler.transform(x_test).

2.3 Feature Selection

Removed Non-Informative Features

The following non-informative features (constant values or unique record identifiers) were removed prior to model development:

- **EmployeeCount** (Constant value = 1)
- **StandardHours** (Constant value = 80)

- **Over18** (*Constant value = 'Y'*)
- **EmployeeNumber** (*Unique ID column*)

2.4 Final Feature Set

- **Dataset Split:** 80% Training Data (x_train, y_train) and 20% Testing Data (x_test, y_test) via train_test_split(test_size=0.2).
- **Input Predictor Matrix (X):** 30 cleaned, capped, encoded, and standardized features:
 - Age, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, MonthlyRate, NumCompaniesWorked, OverTime, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager
- **Target Variable (y):**
 - **Attrition** (Label Encoded: 1 -> Yes, 0 -> No)